Table of transforms

While it's true that we can always use the definition to find the Laplace transform of a function,

$$F(s) = \int_0^\infty e^{-st} f(t)\, dt$$

we don't want to be forced to evaluate this integral again and again, repeating these same steps for the same function. For that reason, we'll often use a table of Laplace transforms for common functions, like the table below.

Table of Laplace transforms

For instance, the first formula in the table tells us that the Laplace transform of the function $f(t) = 1$ is $F(s) = 1/s$.

The third formula in the table tells us that the Laplace transform of $f(t) = t^n$ will be $F(s) = n!/s^{n+1}$, as long as $n = 1,2,3,...$ So if we're trying to transform $f(t) = t^3$, then $n = 3$ and we know from the table that the transform will be

$$\mathscr{L}(t^3) = \frac{3!}{s^{3+1}} = \frac{6}{s^4}$$

Many of the formulas in this table may seem unfamiliar at this point, but we'll get to many of them later on. For now, this will just serve as a resource that we can refer back to as we learn about Laplace transforms.

© Krista King Math



$$\mathscr{L}(f(t)) = F(s)$$

$$\mathscr{L}(1) = \frac{1}{s}$$

$$\mathscr{L}(e^{at}) = \frac{1}{s-a}$$

$$\mathscr{L}(t^n) = \frac{n!}{s^{n+1}} \qquad n = 1, 2, 3, \ldots$$

$$\mathscr{L}(t^p) = \frac{\Gamma(p+1)}{s^{p+1}} \text{ for } p > -1 \qquad \Gamma(t) = \int_0^\infty e^{-x} x^{t-1}\, dx$$

$$\mathscr{L}(\sqrt{t}) = \frac{\sqrt{\pi}}{2s^{\frac{3}{2}}}$$

$$\mathscr{L}(t^{n-\frac{1}{2}}) = \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)\sqrt{\pi}}{2^n s^{n+\frac{1}{2}}} \qquad n = 1,2,3,...$$

$$\mathscr{L}(\sin(at)) = \frac{a}{s^2 + a^2} \qquad \mathscr{L}(\cos(at)) = \frac{s}{s^2 + a^2}$$

$$\mathscr{L}(t\sin(at)) = \frac{2as}{(s^2 + a^2)^2} \qquad \mathscr{L}(t\cos(at)) = \frac{s^2 - a^2}{(s^2 + a^2)^2}$$

$$\mathscr{L}(\sin(at) - at\cos(at)) = \frac{2a^3}{(s^2 + a^2)^2} \qquad \mathscr{L}(\sin(at) + at\cos(at)) = \frac{2as^2}{(s^2 + a^2)^2}$$

$$\mathscr{L}(\cos(at) - at\sin(at)) = \frac{s(s^2 - a^2)}{(s^2 + a^2)^2} \qquad \mathscr{L}(\cos(at) + at\sin(at)) = \frac{s(s^2 + 3a^2)}{(s^2 + a^2)^2}$$

© Krista King Math

$$\mathscr{L}(\sin(at+b)) = \frac{s\sin b + a\cos b}{s^2+a^2}$$

$$\mathscr{L}(\cos(at+b)) = \frac{s\cos b - a\sin b}{s^2+a^2}$$

$$\mathscr{L}(\sinh(at)) = \frac{a}{s^2-a^2}$$

$$\mathscr{L}(\cosh(at)) = \frac{s}{s^2-a^2}$$

$$\mathscr{L}(e^{at}\sin(bt)) = \frac{b}{(s-a)^2+b^2}$$

$$\mathscr{L}(e^{at}\cos(bt)) = \frac{s-a}{(s-a)^2+b^2}$$

$$\mathscr{L}(e^{at}\sinh(bt)) = \frac{b}{(s-a)^2-b^2}$$

$$\mathscr{L}(e^{at}\cosh(bt)) = \frac{s-a}{(s-a)^2-b^2}$$

$$\mathscr{L}(t^n e^{at}) = \frac{n!}{(s-a)^{n+1}}$$

$n = 1, 2, 3, \ldots$

$$\mathscr{L}(f(ct)) = \frac{1}{c}F\left(\frac{s}{c}\right)$$

$$\mathscr{L}(u_c(t)) = \mathscr{L}(u(t-c)) = \frac{e^{-cs}}{s}$$

Heaviside function

$$\mathscr{L}(\delta(t-c)) = e^{-cs}$$

Dirac delta function

$$\mathscr{L}(u_c(t)f(t-c)) = e^{-cs}F(s)$$

$$\mathscr{L}(u_c(t)g(t)) = e^{-cs}\mathscr{L}(g(t+c))$$

$$\mathscr{L}(e^{ct}f(t)) = F(s-c)$$

$$\mathscr{L}(t^n f(t)) = (-1)F^{(n)}(s)$$

$n = 1,2,3,...$

$$\mathscr{L}\left(\frac{1}{t}f(t)\right) = \int_s^\infty F(u)\,du$$

© Krista King Math

$$\mathscr{L}\left(\int_0^t f(v)\ dv\right) = \frac{F(s)}{s}$$

$$\mathscr{L}\left(\int_0^t f(t-\tau)g(\tau)\ d\tau\right) = F(s)G(s)$$

$$\mathscr{L}(f(t+T) = f(t)) = \frac{\int_0^T e^{-st}f(t)\ dt}{1-e^{-sT}}$$

$$\mathscr{L}(f'(t)) = sF(s) - f(0)$$

$$\mathscr{L}(f''(t)) = s^2F(s) - sf(0) - f'(0)$$

$$\mathscr{L}(f^{(n)}(t)) = s^nF(s) - s^{n-1}f(0) - s^{n-2}f'(0)\ldots - sf^{n-2}(0) - f^{n-1}(0)$$

Keep in mind that, given a combination (sum or difference) of functions, we can transform each function individually. In addition, constant coefficients don't affect the value of a transform. We can translate these facts into formulas:

$$\mathscr{L}(af(t)) = aF(s)$$

$$\mathscr{L}(f(t) + g(t)) = F(s) + G(s)$$

Let's do a simple example so that we can see how to start matching values in our function to formulas in the table, and then use the formulas to quickly find the transform of the function.

Example

© Krista King Math

Use a table to find the Laplace transform of the function.

$$f(t) = e^{2t} - \sin(4t) + t^7$$

Because that the transform of a sum is the sum of the transforms, we can transform each term in the function separately. Therefore, our goal will be to find formulas in the table to individually transform e^{2t}, $\sin(4t)$, and t^7.

The transforms we need to use are

$$\mathscr{L}(e^{at}) = \frac{1}{s-a} \qquad \text{with } a = 2$$

$$\mathscr{L}(\sin(at)) = \frac{a}{s^2+a^2} \qquad \text{with } a = 4$$

$$\mathscr{L}(t^n) = \frac{n!}{s^{n+1}} \qquad \text{with } n = 7$$

When we apply these transform formulas to the terms in our function, we get

$$\mathscr{L}(e^{2t}) = \frac{1}{s-2}$$

$$\mathscr{L}(\sin(4t)) = \frac{4}{s^2+4^2} = \frac{4}{s^2+16}$$

$$\mathscr{L}(t^7) = \frac{7!}{s^{7+1}} = \frac{7!}{s^8}$$

Replacing the terms in the original function with their transforms, we can say that the Laplace transform of the function $f(t)$ is

© Krista King Math



$$F(s) = \frac{1}{s-2} - \frac{4}{s^2+16} + \frac{7!}{s^8}$$

© Krista King Math